1 Chapter 4: Electrostatics

1.3 Coulomb's Law

An integral part of any electrostatic analysis is the ability to find the electric field intensity $\vec{E}$ and the associated electric flux density $\vec{D}$ at any given point, based on the influence of some charged point, surface, or volume. (Recall from Chapter 1 that electric field density $\vec{D} = \varepsilon\vec{E}$ and represents the quantity of flux passing through a unit area perpendicular to the field).

Beginning with point charges, Coloumb's Law provides a simple way of deriving the electric field induced by a single point charge with no mass, with a charge of q:

$$\vec{E} = [\frac{q}{4\pi\varepsilon_0 R^2}]\hat{\alpha_R},$$

where R is the radial distance (shortest straight-line path from the charge to the point being measured) and $\hat{\alpha_R}$ is the unit vector pointing away from the charge. The influence of multiple point charges follow the superposition property, and can found as the sum of the electric field intensity due to each individual point charge (this will often require some vector math to dertermine the direction of $\vec{E}$ at some point).

This kind of model is ideal, because it allows us to consider larger distributions of charge, such as charged lines, planes, spheres, and cylinders, as collections of point charges, which we can then integrate to find the sum of their influences (there's an easier way to do this using Gauss's Law, which we'll see shortly).

1.9 Capacitance

He did an example with a coaxial cable at the start but I was running late, just caught the end of it.

Image theory: If a continous charge is present in some region containing an infinite plane of perfectly conductive material, the electric field $\vec{E}$ in the region above the plane may be found by reflecting the source of charge across the conductive plane and flipping it to the opposite polarity, then summing the influence of the two charges in the region above the conductive plane. (This is easier to visualize than explain).

1.10 Review

Using Gauss's Law to find electric field:

Example: Two infinitely long coaxial cylindrical surfaces, $r = a$ and $r = b$, $(b < a)$, carry surface charge densities of ρ_{sa} and ρ_{sb}, respectively.

a) Determine $\vec{E}$ everywhere.

   Use a cylindrical Gaussian surface to find $\vec{E}$ in $r < a$ first. Using Gauss's law,

$$\int_S \vec{E} \cdot d\vec{s} = \frac{Q_{\text{total}}}{\varepsilon_0}.$$

   For this region, $Q_{\text{total}} = 0$, and so there is no electric field $\vec{E}$ in this region.

   In the region $a < r < b$, we first find $\vec{E} \cdot d\vec{s} = \vec{E} \cdot 2\pi rL$, and then $Q_{\text{total}} = \frac{\rho_{sa} \cdot 2\pi aL}{\varepsilon_0}$. Simplifying,

$$\vec{E} = \hat{\alpha_r}\frac{\rho_{sa} \cdot a}{r \cdot \varepsilon_0}\ \frac{\text{V}}{\text{m}}.$$

   In the region $r > b$, we sum the two surface charges, so

$$\vec{E} = \hat{\alpha_r}\frac{\rho_{sb} \cdot a + \rho_{sb} \cdot b}{r \cdot \varepsilon_0}\ \frac{\text{V}}{\text{m}}.$$

   Probably a good idea to memorize the area and volume of cylinders and spheres to quickly calculate total charges as a function of surface/volume charge densities.

Finding electric field and electric flux density using boundary conditions:

Example: Two lossless dielectric regions with $\varepsilon_{r1} = 2$ and $\varepsilon_{r2} = 3$ are separated by the charge free boundary shown below. Find $\vec{E}$ and $\vec{D}$ in both regions given that $\vec{E}_1 = 2\hat{\alpha_x} - 3\hat{\alpha_y} + 5\hat{\alpha_z}$. (Charge free boundary is along the x-y plane.)

Using boundary conditions, first identify tangential and normal components of the $\vec{E}$ field. In the region defined by ε_{r1}, $\vec{E}_{1t} = 2\hat{\alpha_x} + 3\hat{\alpha_y}$ and $\vec{E}_{1n} = 5\hat{\alpha}_z$. Doing the same for the electric flux, $\vec{D}_1 = \varepsilon_1\vec{E}_1 = \varepsilon_0 2(2\hat{\alpha_x} - 3\hat{\alpha_y} + 5\hat{\alpha_z})$.

Applying boundary conditions,

$$E_{1t} = E_{t2} = 2\hat{\alpha_x} - 3\hat{\alpha_y}$$
$$D_{1n} - D_{2n} = \rho_s = 0$$

Solving for D_{2n}, $D_{2n} = 10\varepsilon_0\hat{\alpha_z}$. Then, $E_{2n = \frac{D_{2n}}{\varepsilon_2}} = \frac{10}{3}\hat{\alpha_z}$.

Last homework will involve a question about boundary conditions on a spherical surface.

Reviewed HW 1, question 2 (some people had questions apparently).